# NOGA AHARONY

PhD Candidate in Systems Biology, Columbia University
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Computational biology PhD candidate developing evaluation frameworks and applications for biological language models, with a focus on genome language models, genomic context use, pathway-level reasoning, metagenomic surveillance, and protein-protein interaction prediction. Research spans machine learning for biology, microbial genomics, biosurveillance, and experimental molecular biology.

## EDUCATION

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| **In progress** | **Doctor of Philosophy, Systems Biology**, Columbia University<br>Thesis: Systematic Evaluation Frameworks for Foundational Genome Language Models Development<br>Advisor: Mohammed AlQuraishi, PhD |
| **2021** | **Master of Science, Biology**, Technion - Israel Institute of Technology<br>Thesis: Rapid Gene Content Alteration in Recurring Infections<br>Advisor: Roy Kishony, PhD |
| **2019** | **Bachelor of Science, Honours Neuroscience**, McGill University<br>First Class Honours; Dean's Honours List; Dean's Multidisciplinary Research List |

## RESEARCH EXPERIENCE

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| **2022-Present** | **Mohammed AlQuraishi Lab**, Columbia University, Program of Mathematical Genomics<br>Developing evaluation frameworks and applications for biological language models, especially genome language models, across genomic context use, pathway-level reasoning, wastewater pathogen monitoring, and protein-protein interaction prediction. Co-developing GeneCtx for dissecting genomic context use and WAVEBench for evaluating genomic foundation models under realistic metagenomic distribution shifts. |
| **2021-2023** | **Tal Korem Lab**, Columbia University, Program of Mathematical Genomics<br>Created a model to detect small RNAs in bacterial genomes. |
| **2022** | **Itsik Pe'er Lab**, Columbia University, Department of Computer Science<br>Developed a neural embedding model in hyperbolic spaces to better predict small proteins. |
| **2019-2021** | **Roy Kishony Lab**, Technion - Israel Institute of Technology, Department of Biology<br>Constructed a computational pipeline to identify mobile genetic elements in recurring infections. Designed experiments unraveling eco-evolutionary dynamics of microbial communities. |
| **2018-2019** | **Amine Kamen Lab**, McGill University, Department of Bioengineering<br>Designed and manufactured adeno-associated viruses carrying Cas9 for improved CAR-T cell engineering. |
| **2018** | **Center for Health Security**, Johns Hopkins University, School of Public Health<br>Surveyed DNA synthesis providers' sequence- and customer-screening practices; this work informed the Health Security governance paper *Strengthening Security for Gene Synthesis*. Studied safety, innovation, and community norms in the DIY biology community. |
| **2017-2018** | **Jonathan Kimmelman Lab**, McGill University, Biomedical Ethics Unit<br>Applied machine learning to analyze clinical trial design factors that influence forecasts of outcomes; contributed to a meta-analysis. |
| **2017** | **Edward Ruthazer Lab**, McGill University, Montreal Neurological Institute<br>Electroporated and imaged tadpoles to study structural plasticity in the developing visual system. |
| **2016-2017** | **Rafael Najmanovich Lab**, University of Montreal, Department of Pharmacology and Physiology<br>Developed software characterizing protein interfaces based on surrounding force fields. Modeled biophysical features of *C. difficile* germination protease to identify candidate inhibiting ligands. |

## PUBLICATIONS AND MANUSCRIPTS

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| **2026** | Le Lievre T*, **Aharony N***, Wang S, Adams E, AlQuraishi M. GeneCtx: An Evaluation Framework for Dissecting Genomic Context Use in Genome Language Models. Submitted to NeurIPS 2026 Evaluations & Datasets track. |
| **2023** | Liang C, Wagstaff J, **Aharony N**, Schmit V, Manheim D. Managing the Transition to Widespread Metagenomic Monitoring: Policy Considerations for Future Biosurveillance. *Health Security*, 21(1), 34-45. |
| **2022** | Myers A, Utpala S, Talbar S, Sanborn S, Shewmake C, Donnat C, Mathe J, Lupo U, Sonthalia R, Cui X, Szwagier T, Pignet A, Bergsson A, Hauberg S, Nielson D, Sommer S, Klindt D, Hermansen E, Vaupel M, Dunn B, Xiong J, **Aharony N**, Pe'er I, Ambellan F, Hanik M, Nava-Yazdani E, von Tycowicz C, Miolane N. ICLR 2022 Challenge for Computational Geometry & Topology: Design and Results. arXiv. |
| **2021** | Milman O*, Yelin I*, **Aharony N**, Katz R, Herzel E, Ben-Tov A, Kuint J, Gazit S, Chodick G, Patalon T, Kishony R. Community-level evidence for SARS-CoV-2 vaccine protection of unvaccinated individuals. *Nature Medicine*. |
| **2020** | Yelin I*, **Aharony N***, Shaer-Tamar E*, Argoetti A*, Messer E, Berenbaum D, Shafran E, Kuzil A, Gandali N, Hashimshony T, Mandel-Gutfreund Y, Halberthal M, Geffen Y, Szwarcwort-Cohen M, Kishony R. Evaluation of COVID-19 RT-qPCR test in multi-sample pools. *Clinical Infectious Diseases*. |
| **2019** | Moco PD, **Aharony N**, Kamen A. Adeno-Associated Viral Vectors for Homology-Directed Generation of CAR-T cells. *Biotechnology Journal*. |

* Equal contribution.

## POSTERS AND TALKS

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| **2026** | Le Lievre T*, **Aharony N***, AlQuraishi M. PRIME-Bench: A Benchmark for Pathway-Level Reasoning in Genome Language Models. Accepted poster, ISMB 2026. |

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| **2026** | **Aharony N**. Who Holds the Reins in Agent-Assisted Systems Biology? Accepted poster and extended abstract, NE AI Agents Day 2026. |
| **2026** | **Aharony N**, AlQuraishi M. Evaluating Genome Language Models for Metagenomic Binning. Accepted poster, Pioneering Israeli Science Conference / ScienceAbroad 2026. |
| **2026** | **Aharony N**. Rapid Talks and Panel: 2024 Innovative Tools & Solutions Award Recipients. Flash talk and panelist, PRI. |
| **2025** | **Aharony N**. Genome Language Models Reading the Future of Pandemics in Wastewater. Flash talk, PRI Innovation Showcase. |
| **2022** | **Aharony N**. Computational Methods for Pandemic Pathogen Detection. Flash talk, Singer Symposium. |
| **2021** | **Aharony N**, Kishony R. Rapid Alteration in Genetic Content Upon Recurring Infection. Poster, EMBL Symposium: New Approaches and Concepts in Microbiology. |
| **2020** | **Aharony N**. Careers in Biosecurity. Flash talk, EAGx Virtual. |
| **2018** | **Aharony N**, Gronvall G. How Secure is the Gene Synthesis Industry? Poster, Biological Weapons Convention: Meeting of Experts. |

## AWARDS AND FUNDING

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| **2024** | NYC-PRI Innovative Tools and Solutions to Address Public Health Emergencies. |
| **2022** | EA Funds PhD Support Scholarship. |
| **2021** | Open Philanthropy Early-Career Funding. |
| **2020** | Miriam and Aaron Gutwirth Memorial Fellowship for Excellence in Research. |
| **2019** | Leonard and Diane Sherman Interdisciplinary Graduate School Fellowship. |
| **2018** | Open Philanthropy Early-Career Funding for Global Biological Risks. |
| **2017, 2018** | McGill University Faculty of Science Scholarship. |
| **2017** | CIHR Undergraduate Research Award in Computational Biology. |
| **2016** | NSERC Undergraduate Student Research Award; FRQNT Top-Up to NSERC Undergraduate Student Research Award. |

## TEACHING, MENTORSHIP, AND SCIENTIFIC SERVICE

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| **2016-2018** | **Teaching Assistant**, McGill University<br>Created problem sets, held office hours, and led review sessions for biochemistry, molecular biology, neuroscience, and introductory physics courses. |
| **2018-2023** | **Research and career mentorship**<br>Mentored students and early-career professionals interested in biosecurity and computational biology projects through Effective Thesis Project and 80,000 Hours referrals. |
| **2019** | **Volunteer Consultant**, Battelle Memorial Institute / ThreatSEQ<br>Consulted on a ThreatSEQ joint initiative with the FBI assessing whether gene synthesis companies would detect orders containing genes from the Select Agents list. |

## SKILLS

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| **Computational** | Python, R, MATLAB, C/C++, Java; biological language model evaluation; sequence modeling; metagenomic analysis; hyperbolic embeddings; genomic pipeline development. |
| **Experimental** | Microbial evolution experiments; AAV vector production; CRISPR/Cas9 delivery; molecular biology workflows. |
| **Languages** | Hebrew (native), English (near-native), French (novice). |